09/07/2026

Instructions for using the SNP_GWAS software

T. Tribout, D. Boichard

Université Paris Saclay, INRAE, AgroParisTech, GABI, Jouy-en-Josas, France.

Version: SNP_GWAS V1.1

This documentation is licensed under the Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0).
You are free to share and adapt this documentation, provided that appropriate credit is given to the author and that the material is not used for commercial purposes.
The full text of the license is available at: https://creativecommons.org/licenses/by-nc/4.0/

Introduction

The SNP-GWAS software is a program developed in INRAE, enabling GWAS to be carried out on a large number of individuals with a phenotype and genotypes (up to hundreds of thousands of individuals), and allowing to jointly apply various options that are not found simultaneously in other available software (use of allelic dosages instead of discrete genotypes, phenotype weighting, consider a dominance effect in addition to the usual additive effect for the tested variant, …).

NB: This program requires an implementation of the BLAS and LAPACK libraries. It has been developed and tested with Intel oneMKL but should also work with other compatible implementations such as OpenBLAS.

Naming conventions

In the following, we'll call:

- "variant" or "tested variant" (**TV**) the biallelic variant tested in GWAS and whose effect(s) on phenotype we want to estimate and test;

- "markers" or "SNP" the biallelic markers spread across the genome considered to calculate the genomic relationships and for which we're not interested in the individual effect on phenotype;

- « Phenotypes » or « performances » the numeric records of the individuals = entries of vector *y* in the following models;

- « weight » of the phenotype of the individual i the w_i value such that the residual e_i of y_i from the GWAS model is $\aleph(0,\frac{\sigma_e^2}{w_i})$, with σ_e^2 being the residual variance of the considered trait; w_i is similar to a number of own phenotypes for animal i. The W matrix is the diagonal n x n matrix such as W(i,i)= w_i.

General model used in the program

Usually, the model considered to describe the vector y of phenotypes in GWAS is the following:

$$y = X\beta + Tu + g_k a_k + e \qquad (1)$$

where

y is a $n \times 1$ vector of phenotypes with n being the sample size (i.e. the number of individuals with both phenotype and genotypes),

β is a vector of fixed effects including *a minima* an overall mean,

X is the incidence matrix relating β to y,

u is a $n \times 1$ vector of the polygenic breeding values of the individuals assumed to be normally distributed as $u \sim \aleph(0, K\sigma_u^2)$, with σ_u^2 being the polygenic genetic variance of the considered trait and K being a kinship matrix relating individuals to each other,

T is the incidence matrix relating u to y,

a_k is the fixed effect of the k[th] tested biallelic variant on the phenotype, considered in the model as a fixed regression coefficient whose covariates are in the $n \times 1$ vector g_k,

e is the $n \times 1$ vector of residuals assumed to be normally distributed as $e \sim \aleph(0, I_n\sigma_e^2)$, with I_n being a n x n identity diagonal matrix and σ_e^2 being the residual variance of the considered phenotype.

The mixed model equations corresponding to this usual model are then:

$$\begin{bmatrix} X'X & X'T & X'g_k \\ T'X & T'T + \rho K^{-1} & T'g_k \\ g'_kX & g'_kT & g'_kg_k \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{u} \\ \hat{a}_k \end{bmatrix} = \begin{bmatrix} X'y \\ T'y \\ g'_ky \end{bmatrix}, \qquad \text{with } \rho = \frac{\sigma_e^2}{\sigma_u^2} \qquad (2)$$

In particular, we note that the size of the complete system (2) increases linearly with the number of individuals in the sample, and solving these equations to calculate the variant estimated effect(s) and PEV requires being able to obtain the inverse of the K relationship matrix, which can be an extremely time-consuming task if K is based on genomic relationships (and therefore is dense) and for large samples.

In SNP-GWAS software, model (1) based on a genomic relationship matrix was equivalently reformulated as:

$$y = X\beta + Ms + g_ka_k + e\,, \qquad (3a)$$

or more generally

$$y = X\beta + Ms + Z_kb_k + e\,, \qquad (3b)$$

where

$b_k = a_k$ and $Z_k = g_k$ when only the additive effect a_k of the k[th] biallelic variant is considered, or $b'_k = [a_k \quad d_k]$ and $Z_k = [g_k \quad \delta_k]$ when both additive effect a_k and dominance effect d_k of the k[th] biallelic variant are considered *(see below for the meaning and values of g_k and δ_k covariates)*,

and $Ms = Tu$ is the n x 1 vector of polygenic breeding values of individuals in the sample,

with s is an $n_s \times 1$ vector of SNP effects with $s \sim \aleph(0, I_s\sigma_s^2)$, I_s is an $n_s \times n_s$ identity matrix and $\sigma_s^2 = \frac{\sigma_u^2}{\sum_{j=1,n_s} 2p_j(1-p_j)}$, where p_j is the frequency of the second allele of the j[th] SNP, σ_u^2 is the genetic variance of the considered phenotype (considered known),

M is an $n \times n_s$ matrix of centered genotypes whose (i,j) element is $m_{ij} = (g_{ij} - 2p_j)$, where g_{ij} is the number of copies of allele 2 for the j[th] SNP carried by the i[th] individual, b_k is the $n_b \times 1$ vector of fixed additive effect and possibly dominance effect of the k[th] TV,

The mixed model equations corresponding to the model (3b) are then:

$$\begin{bmatrix} X'X & X'M & X'Z_k \\ M'X & M'M + \rho I_s & M'Z_k \\ Z'_kX & Z'_kM & Z'_kZ_k \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{s} \\ \hat{b}_k \end{bmatrix} = \begin{bmatrix} X'y \\ M'y \\ Z'_ky \end{bmatrix}, \qquad \text{with } \rho = \frac{\sigma_e^2}{\sigma_s^2} \qquad (4a)$$

We can see that the size of this system of equations is independent of the number of individuals in the dataset but depends on the number of markers considered, and the inverse of the kinship matrix does not appear directly anymore. A few tens of thousands of informative markers being sufficient to model polygenic genetic value u, it is therefore possible to carry out GWAS on datasets of several million individuals at a limited memory cost.

Of course, the set of n_s biallelic markers considered to model the breeding values and the set of tested TV can be different. A typical situation is to use a set of 15,000 to 40,000 markers to model genetic polygenic values and to test successively the effect of several hundred thousand to several million variants per chromosome.

Different weight for individual's phenotypes. As a general rule, an individual's phenotype y_i is its performance pre-adjusted for environmental effects (sex, batch, season, ..), or a weighted average of its adjusted performances (such as a YD) if the measurement is repeated several times in the animal's life, or an average of the adjusted performances of a sire's daughters (such as a DYD); in the latter two cases, the phenotype is then accompanied by a weight ω_i corresponding to the equivalent number of performances of the individual.

In this case, the vector e in model 3b is an $n \times 1$ vector of residuals such as $e \sim \aleph(0, W^{-1}\sigma_e^2)$, where W is a $n \times n$ diagonal matrix with (i,i) element is the weight ω_i of i[th] individual's phenotype y_i and σ_e^2 is the residual variance of the phenotype (considered known).

Then, the corresponding MME are:

$$\begin{bmatrix} X'WX & X'WM & X'WZ_k \\ M'WX & M'WM + \rho I_s & M'WZ_k \\ Z'_kWX & Z'_kWM & Z'_kWZ_k \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{s} \\ \hat{b}_k \end{bmatrix} = \begin{bmatrix} X'Wy \\ M'Wy \\ Z'_kWy \end{bmatrix}, \qquad \text{with } \rho = \frac{\sigma_e^2}{\sigma_s^2} \qquad (4b)$$

In the case where the phenotypes are unweighted, i.e. all ω_i = 1, *W* = *I*.

Naming conventions

In the following, we will call "LHS" and "RHS", respectively, the left hand side and the right hand side of the Mixed Model Equation system (4b) reduced by the equations for the variant effect(s), i.e. the part of the model common to all TV:

$$\begin{bmatrix} X'WX & X'WM \\ M'WX & M'WM + \rho I_s \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{s} \end{bmatrix} = \begin{bmatrix} X'Wy \\ M'Wy \end{bmatrix} \qquad (5)$$

$$LHS = \begin{bmatrix} X'WX & X'WM \\ M'WX & M'WM + \rho I_s \end{bmatrix} \qquad RHS = \begin{bmatrix} X'Wy \\ M'Wy \end{bmatrix}$$

Program functionalities

In its current version (**SNP_GWAS**), the SNP-GWAS software can perform GWAS:

- considering
  - allelic dosages at TV, varying from 0 to 2;
  - discrete genotypes equal to the number of allele "2" (0, 1 ou 2) carried by individual at TV;
- estimating and testing for the TV
  - (1) its additive effect;

    - (2) its additive effect and its dominance effect (when the data make it possible);
- considering unweighted (default) or weighted phenotypes;
- possibly considering several environmental fixed effects (categorial effects or regression coefficients), for example principal components to account for population structuring;
- considering
    - the set of markers specified by the user;
    - a set of n_s markers evenly distributed across the genome and the most informative selected by the program from a larger set of markers supplied by the user;
- using or excluding the markers of the chromosome carrying the TV (similar to LOCO = Leave One Chromosome Out option in GCTA), or excluding the markers of a specified length sliding segment of the chromosome carrying the TV ("leave one Segment out");
- fitting the additional additive effect of a user specified variant as a fixed effect in the GWAS model, to estimate and test the joint remaining effect of TV on the same chromosome;

<u>**Limitations and requirements**</u>

- The non-genetic part of the model must include at least one fixed effect (e.g. an overall mean), which must be explicitly specified by the user;
- one unique phenotype per individual (no repeated phenotypes);
- single trait GWAS;
- the residual variance σ_e^2 and the genetic variance σ_u^2 considered in the MME are supposed known, and should therefore be previously estimated on the data with the same model (excluding TV), breeding value based model or Marker based model, since they are equivalent;
- Each individual is present <u>at most once</u> in the phenotype file, the marker genotype file, the TV genotype / allelic dosage file (no duplicate); **the GWAS are performed considering the individuals with a phenotype.** Marker genotype file and Variant genotype/allelic dosage file can have individuals not present in the phenotype file, **but all individual present in the phenotype file have to be present in the Marker genotype file and in the Variant genotype/allelic dosage file**. Presence of individuals in Phenotype file that are missing in Marker Genotype / Allelic Dosage file or missing in Variant Genotype file will result in stopping the program.
- Individual identifiers can be alphanumeric, but cannot exceed 20 characters in length;
- The program performs GWAS for one single trait; if several traits have to be considered, all steps must be completed from the beginning for each trait;
- The software has been developed for (very) large populations, but can obviously handle small datasets, even if Marker Based model may result in longer computation times than Breeding Value based model in this case. If the number of individuals is smaller than the number of markers and if speed of execution is crucial, use another program.

Successive steps in a GWAS with SNP-GWAS

GWAS analysis of chromosome variants is carried out in 3 successive steps using the same program, as shown below.

If GWAS is to be performed on several chromosomes on the same population, the first step needs to be performed once, and only steps 2 and 3 need to be repeated for each chromosome.

- **Step 1: program SNP_GWAS / performed once whatever the number of chromosomes with TV**
  - Performance file reading;
  - Marker genotype file reading;
  - Marker map file reading;
  - If needed, exclusion/selection of markers to be considered in the GWAS, according to user's specifications in parameter file (MAF, number of markers, ..), and writing of a map file for markers useful for next steps → creation of a new marker map file with updated keep/remove indicator for markers;
  - LHS building and inversion ; LHS^{-1} writing to disk in binary format;
  - RHS building and writing to disk in binary format.

- **Step 2: program SNP_GWAS / performed for each chromosome carrying TV**
  - TV genotype / allelic dosage file reading;
  - TV map file reading;
  - Marker genotype file reading;
  - (useful) marker map file reading;
  - According to user's options, creation of groups of TV that will be analyzed together, determining for each group which markers, if any, will be excluded from the equations;
  - Creating a directory tree on disk (1 folder per TV group), and write the following files to these directories:
    - 1 binary genotype / allelic dosage file for the TV of the group (1 line per TV, allowing in the following step to treat the TV 1 by 1) if the user provides genotype/allelic dosage file where TV are in columns ;

      or

      1 binary file for the TV of the group containing the Id of the TV of the group if the user provides minimac4 VCF allelic dosage files (several VCF files containing the same TV can be provided, each containing allelic dosages for different animals);
    - 1 file specifying the 1st and last marker, if any, to be excluded from the equations for the TV group;
    - 1 parameter file read by SNP-GWAS program to perform the 3rd step on the TV group;
  - Writing in the root working directory of a file specifying the characteristics of each TV group (group ID, TV number, excluded variants, …);
  - Writing in the root working directory of a file containing the command lines to launch step 3 for all the groups in parallel on a cluster.

- **Step 3: program SNP-GWAS/ performed for each TV group defined at Step 2 of each chromosome carrying TV**
  - Performance file reading;
  - Marker genotype file reading;
  - (useful) marker map file reading;
  - binary genotype / allelic dosage file for the TV reading (line by line = TV by TV), or minimac4 VCF files reading (line by line = TV by TV);
  - TV map file reading;

- Binary files containing LHS^{-1} and RHS created at Step 1;
- If needed, LHS^{-1} and RHS adjustment by removing the equations for markers to be removed for the TV group;
- For each TV in the group:
  - Estimation of TV effect(s) requested by the user;
  - Calculation of the variance of the model residuals, including the effect of the considered TV;
  - Calculation of the Prediction Error (Co)Variances for the considered TV effect(s);
  - Calculation of the test-statistics for the considered TV effect(s);
  - Calculation of the P-value associated with the test statistic for the considered TV effect(s);
  - On disk writing of the results for the TV in its variant group result file.

- **Step 4 : program Cumul_Res_SG**
  Once Step 3 has been completed for all the TV groups for the chromosome:
  - Reading of all the TV group result files;
  - Calculation of -LOG10(P-value) for each estimated effect of each TV;
  - Write results for all TV from all groups in a unique result file, ordered according to TV position on the chromosome.

If GWAS are to be carried out for several traits in several breeds, the following directory organization is proposed:

/read-only directory containing data :	performance files (different files for different breeds), marker genotype files (different files for different breeds), marker map files, TV genotype/allelic dosage files (different files for different breeds), TV map files, SNP-GWAS parameter files (1 param file per breed x trait x TV chromosome), …
/work_folder/…/BREED1/TR1/GWAS :	working directory for TRAIT1 in BREED1: program launch, log writing, … ; the chromosomes x TV groups subfolders for this breed and this trait will be created in this working directory in step 2;
/work_folder/…/BREED1/TR1/binfiles :	the LHS^{-1} and RHS binary files for BREED1 and TRAIT1 will be written here in step 1;
/work_folder/…/BREED1/TR2/GWAS :	working directory for TRAIT2 in BREED1: program launch, log writing, … ; the chromosomes x TV groups subfolders for this breed and this trait will be created in this working directory in step 2;
/work_folder/…/BREED1/TR2/binfiles :	the LHS^{-1} and RHS binary files for BREED1 and TRAIT2 will be written here in step 1;
…	
/work_folder/…/BREED2/TR1/GWAS :	working directory for TRAIT1 in BREED2: program launch, log writing, … ; the chromosomes x TV groups subfolders for this breed and this trait will be created in this working directory in step 2;
/work_folder/…/BREED2/TR1/binfiles :	the LHS^{-1} and RHS binary files for BREED2 and TRAIT1 will be written here in step 1;
…	

The contents of the parameter files for steps 1 to 4 and how to fill them in are described on the following pages on the basis of a simple model with only an additive effect for the variant, and an overall mean for the non-genetic effects part: $y = \mu + Ms + g_k a_k + e$

Note that the parameter file for Step_3 is automatically created by the program during Step_2.

Then, the use of models with additional effects is explained in later sections.

NB: some options (end of the parameter files for Steps 1 to 3, specified through lines as: “OPTION keyword value”) are only useful for a given step. However, it is simpler to put all the options that will be used in any of the GWAS steps in parameter files for Steps 1 and 2.

Moreover, options specifically useful for Step_3 have to be provided in the parameter file for Step_2 (no problem to put them in parameter file for Step_1 too), since parameter file for Step_3 is created by the program in Step_2 on the basis of the information provided for Step_2.

Use of the program for a GWAS with a basic model

The following parameter files correspond to a GWAS performed with the following characteristics:

- Use of discrete genotypes (0 = genotype 11 ; 1 = genotype 12 or 21 ; 2 = genotype 22) for the TV;
- The TV are on chromosome 14;
- The list of markers to model the animals' breeding values is provided by the user;
- The only non-genetic effect considered in the model is an overall mean;
- Only the additive effect of the TV is estimated and tested;
- No marker is removed from analysis (neither the markers on the chromosome carrying the TV, nor any marker on a segment of the chromosome carrying the TV), except Markers whose MAF is lower than the minimum MAF specified by default or by the user in parameter file;
- Each group of TV in step 3 contains 10000 TV.

Parameter files are text files constituted by successions of KEYWORDS followed by `information on the next line`; **order of KEYWORDS can not be changed in parameter files.**

> The parameter files for step 1 and step 2 have to be prepared by the user.
>
> The parameter file for step 3 and the executable file for launching step 3 will be automatically prepared by the software during step2.

Some options are specific to a given step, but all options can be specified in the parameter file for any step (so, it is convenient tu put all the options in the parameter file as soon as Step 1).

Parameter file read by SNP-GWAS at Step 1:

This parameter file for Step 1 has to be prepared by the user, for each of the chromosome he wants to analyze.

NB: the information in red are comments and should not be present in the parameter file!

```
# Example param file for STEP 1 prg SNP_GWAS      1 line for title
STEPS
1                         ← number of the Step that will be executed (1 for Step 1; 2 for Step 2; 3 for Step 3)
EXEC_PATH
/softwares/SNP_GWAS/bin/SNP_GWAS     ← path for binary of the program, needed in Step2 for STEP 3
BIN_DIR
/work/DIR2/Example3/binfiles     ← folder where binary files for LHS-1 & RHS will be written
DATAFILE
/work/DIR2/Example3/FILES/data_file.txt     ← performance file
NUMBER_OF_EFFECTS
1                ← number of non-genetic effects in the model (>= 1), at least 1 for the overall mean
POS_ID_CHAR
6                 ← column number of animal Id in performance file: always last column read!
OBSERVATION(S)
4              ← column number of phenotype in performance file
WEIGHT(S)
             ← column number of phenotype's weight in performance file (leave empty if no weight, i.e. weight=1)
```

```
EFFECTS: POSITIONS_IN_DATAFILE NUMBER_OF_LEVELS TYPE_OF_EFFECT  [EFFECT NESTED]
1 1 cross   ← column number of first non-genetic effect in performance file / number of levels / effect type
RESIDUAL_VARIANCE
18.76608   ← residual variance of the trait for a phenotype whose weight is 1= $\sigma_e^2$, supposed known
GENETIC_VARIANCE
6.06204     ← genetic variance of the trait = $\sigma_u^2$, supposed known
TYP_PAR_FILE
/work/DIR2/Example3/FILES/Markers_50K.raw   ← marker genotype file
FORMAT_TYP_PAR
plink                ← format type of marker genotype file  (among plink, typ_eval)
MAP_PAR_FILE
/work/DIR2/Example3/FILES/Markers_50K.map   ← marker map file
TYP_GWAS_FILE
                                              ← TV genotype file
FORMAT_TYP_GWAS
                     ← format type of TV genotype/allelic dosage file  (among minimac, typ_eval, plink)
NUMCHR_GWAS
                                          ← chromosome carrying the TV
MAP_GWAS_FILE
                                                ← TV map file
GWAS_TYPE (add / add_dom)
                                        ← effect(s) to be considered in the model for TV
SEGM_REM
                                     ← which markers will be removed form equations during GWAS
COJO
              ← 0 if no variant fit in the model as fixed effect; 1 if 1 variant fit in the model as fixed effect*
OPTION missing -9999.0            ← OPTIONS , see below for more details on OPTIONS
OPTION MAF_minimum 0.20
OPTION NbMaxVarGWAS 15000
OPTION CalcVarRes optim
OPTION seuil_test1 2.0
OPTION StepFichOut 100
OPTION SelAutoSNPpar 10000
```

EXEC_PATH

The « path for the binary" of SNP-GWAS is not needed to execute Step 1, but is needed to prepare the command file to launch Step 3. Not elegant, but it works …

POS_ID_CHAR

The POS_ID_CHAR value is the number of the column in the performance file containing the animal's Id; this must be the last column that is read by the program in Performance file: all the other variables are numeric (phenotype, effect cells, covariates) and have to be before ID in the performance file, only ID is read as alphanumeric. Again, not elegant, but it works…

DATAFILE

After this keyword, give the name of the performance file.

In performance file, fields are separated by 1 or several spaces.

This file contains 1 line per animal, containing numeric data corresponding to

- The levels of fixed effects and covariates for fixed regression coefficients considered in the model (for an overall mean, put a covariate equal to 1 for each individual); for categorial effects with several levels, the levels have to be coded from 1 to N;
- The phenotype for the considered trait (missing phenotype is possible);
- If needed, the weight of this phenotype.

NB : at least one fixed effect has to be included in the model. If the user does not want to include any fixed effects (for example, if phenotypes have already been adjusted for environmental effects), an overall mean must be explicitly added to the model. **To do this, the performance file must contain a column of 1**, which will allow a single-level fixed effect (*i.e.* the overall mean) to be specified in the parameters file.

The Performance file can contain numeric fields that are not read for this GWAS, such as phenotypes and weight for other traits that will be considered in another GWAS.
The last field (from left to right) of the individual's line read by the program is the animal's Identifier, possibly alphanumeric (**max length =20**).

NB: The Performance file can contain values on the right of the animal's ID, but these fields can't be read by the program.

Example of performance file:

```
1   3   2    124   0.92       2          0.9     FR2345
1   2   1    138   0.98       7          1        FR5432
1   1   3    163   0.99        5         1        FR34334
1   2   5    127   0.74   -9999.0      0        FR15313
1   1   4    156   0.82       12        0.8     FR42211
```

field #1 = effect level for overall mean (equal to 1 for all individuals)
field #2 = effect level for a fixed categorial effect with 3 levels
field #3 = not used in this GWAS
field #4 = phenotype for trait 1 for which GWAS will be performed
field #5 = weight of the phenotype for trait 1 for which GWAS will be performed
field #6 = phenotype for trait 2, **not used here** ; animal FR15313 has missing phenotype
field #7 = weight of phenotype for trait 2, **not used here**
field #8 = alphanumerical Id of the individual.

EFFECTS: POSITIONS_IN_DATAFILE NUMBER_OF_LEVELS TYPE_OF_EFFECT [EFFECT NESTED]
In this example, the first and only non-genetic effect is the overall mean, whose level number (= 1 for all individuals) is given in column #1 in performance file; the type effect for this effect is "cross" = categorial fixed effect.
Possible types for effects are "cross" for categorial fixed effects, and "cov" for regression coefficients.
Levels in Performance file for categorial fixed effects have to be numeric, numbered from 1 to N.

RESIDUAL_VARIANCE & GENETIC_VARIANCE

After these keywords, give, respectively, the residual variance σ_e^2 and the total genetic variance σ_u^2 of the trait, such that the heritability of the trait in the GWAS sample population is $h^2 = \sigma_u^2/(\sigma_u^2 + \sigma_e^2)$. A simple way of knowing these variance components is to estimate them with a program like REMLF90, AIREMLF90 or WOMBAT on the files prepared for GWAS with the model corresponding to the system of equations (5) presented above.

TYP_PAR_FILE
After this keyword, give the Marker genotype file, that contains the genotypes for markers of all chromosomes in one single file. **In this file, the markers (1 Marker per row) are ordered by ascending chromosome, and then ascending position within a chromosome.**
The genotypes for a marker are the number of "2" alleles carried by the individual at the marker:
Homozygous 11: genotype = 0
Heterozygous 12 or 21: genotype = 1
Homozygous 22: genotype = 2
Unknown genotypes have to be coded 5; the unknown genotypes will be replaced in the equations by the most frequent genotype observed in the sample for this marker. Therefore, the markers with lots of unknown

genotypes have to be removed from the marker genotype file (or have to be indicated to be removed in the marker map file, see below) by the user.

FORMAT_TYP_PAR

After this keyword, indicate the format for the Marker genotype file among the 2 formats of files that can be read by the program:

typ_eval: 1 single line per individual, starting with the individual Id, 1 or several spaces (**no tab**), followed by its genotypes (numbers of allele "2" (0 1 2) carried by individual) for all markers ordered in ascending chromosome and ascending position on chromosome, without any separation. The first position of the genotype of the first marker has to be the same for all individuals, whatever the individual's Id length. No header.

```
FRXXX8037618    02121212110222
FRXXX9018       12121211220211
FRXXX001395     21022221110220
FRXXX0035018    11220210112222
```

plink: **.raw** file created by **plink software**, **with NA for unknown genotypes replaced by 5 beforehand** (SNP_GWAS reads genotypes as numerics); header line:

NB: FID (family ID) and IID (individual ID) have to be the identical; PAT, MAT, SEX, PHENOTYPE are not used.

```
FID IID PAT MAT SEX PHENOTYPE F0100190_A D5010001_A F0100220_A RGX1000_A RGX2000_A RGX1001_A …
XXXXX000002130 XXXXX000002130 XXXXX4598225 XXXXX2754989 0 -9 2 0 1 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 0
XXXXX000002131 XXXXX000002131 XXXXX2598225 XXXXX2754989 0 -9 2 0 1 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 0
XXXXX000002132 XXXXX000002132 XXXXX2598225 XXXXX2754989 0 -9 2 0 1 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 0
XXXXX000002133 XXXXX000002133 XXXXX2598225 XXXXX2754989 0 -9 1 0 1 0 0 0 0 0 0 1 1 0 0 0 1 0 0 2 1 0
XXXXX000002134 XXXXX000002134 XXXXX2598225 XXXXX2754989 0 -9 1 0 2 0 0 0 0 0 0 0 0 0 0 0 1 0 0 1 1 0
XXXXX000002135 XXXXX000002135 XXXXX2598225 XXXXX2754989 0 -9 1 0 2 0 0 0 0 0 0 0 0 0 0 0 1 0 0 1 1 0
```

Important: For both typ_eval and plink marker genotype files, unknown genotype at a marker is coded 5. PLINK software sets unknown genotypes at 'NA' in .raw file, the user has to replace the NA by 5 before running SNP_GWAS software.

MAP_PAR_FILE

After this keyword, give the name of the map file for the markers present in the marker genotype file.
The map file has to be ordered in ascending chromosomes and in ascending position on chromosome. The map file has to contain the same markers as the genotype file in the same order; 1 line per marker.

If FORMAT_TYP_PAR = **plink**: **.map** file created by **plink software**: (col#1 = chromosome; col#2=marker name; col#3=?; col#4 = position on chromosome).

```
1       F0100190        0       776231
1       D5010001        0       904701
1       F0100220        0       907810
1       RGX1000 0       989141
1       RGX2000 0       990551
1       RGX1001 0       990831
1       F0133440        0       993060
…
```

NB: in this case, the user can add a 5th column in the plink .map file to indicate for each marker if it should be considered (1) or excluded (0) when building the equations:

```
1       F0100190        0       776231 1
1       D5010001        0       904701 1
1       F0100220        0       907810 0
1       RGX1000 0       989141 0
1       RGX2000 0       990551 1
1       RGX1001 0       990831 1
1       F0133440        0       993060 0
…
```

If this 5th column is missing, the program considers that all markers have to be considered (except if optional minimum MAF or automatic choice of a subset of markers is requested by the user).

If FORMAT_TYP_PAR = **typ_eval**: the user has to prepare a marker map file containing 4 columns:

Col#1 = chromosome; col#2 = position of the marker on chromosome (bp); col#3 = order of the marker in the file (from 1 to N, all chromosome confounded); col#4 = indicator for each marker telling if it should be considered (1) or excluded (0) when building the equations:

```
1 776231 1 1
1 904701 2 1
1 907810 3 0
1 989141 4 0
1 990551 5 1
1 990831 6 1
1 993060 7 0
…
```

NB: a marker whose last column in map file has value 1 for considered/excluded indicator will still be removed form the equations if its MAF is lower than the minimum MAF considered by the program (see below).

TYP_GWAS_FILE

This is the keyword for giving the name of the Variant genotype/allelic dosage file. This file is not read in Step 1, but the keyword must be in the parameter file followed by an empty line (or a line filled in with something that won't be used in Step 1, e.g. put the TV genotype/allelic dosage file for one of the chromosomes that will be considered in the following steps).

FORMAT_TYP_GWAS

This is the keyword for giving the format of the genotype/allelic dosage file for variant present in the Variant genotype/allelic dosage file. This file is not used in Step 1, but the keyword must be in the parameter file followed by an empty line (or a line filled in with something that won't be used in Step 1).

NUMCHR_GWAS

This is the keyword to indicate the number of the chromosome carrying the variants for which effects will be estimated and tested in the following steps. Not used in Step 1, but the keyword must be in the parameter file followed by an empty line (or a line filled in with something that won't be used in Step 1).

MAP_GWAS_FILE

This is the keyword to indicate the name of the map file for the variant present in the genotype/allelic dosage file. Not used in Step 1, but the keyword must be in the parameter file followed by an empty line (or a line filled in with something that won't be used in Step 1).

SEGM_REM

This is the keyword for specifying the strategy retained for removing markers on the same chromosome as the variants in next steps. Not used in Step 1, but the keyword must be in the parameter file followed by an empty line (or a line filled in with something that won't be used in Step 1).

COJO

This is the keyword for specifying the marker, if any, to fit in the model while estimating and testing the effect of the variants in following steps (joint analysis). Not used in Step 1, but the keyword must be in the parameter file followed by an empty line (or a line filled in with something that won't be used in Step 1).

Output files created by Step_1:

During Step_1, common step for all chromosomes that will be subsequently analyzed in Step_2 and Step_3 for the same trait, SNP_GWAS program creates on disk, ***in the directory indicated after the keyword*** ***`BIN_DIR`,*** the following files:

- 1 binary file named **invLEFT_binfile** and 1 binary file **RIGHT_binfile** containing, respectively, the inverse of the left hand side and the right hand side of the system without any variant:

$$\begin{bmatrix} X'WX & X'WM \\ M'WX & M'WM + \rho I \end{bmatrix} \begin{bmatrix} \beta \\ \alpha \end{bmatrix} = \begin{bmatrix} X'My \\ M'Wy \end{bmatrix}$$

- In case the user used « OPTION SelAutoSNPpar NNN » to request the program to select automatically the most informative markers with MAF > MAF_minimum in NNN segments in the genome among the Markers in TYP_PAR_FILE and MAP_PAR_FILE, Step_1 creates a file names **carte_SNP_PAR_bis.txt** located in the working directory, that will be used in following steps to indicate the markers to be considered in the equations.

  This **carte_SNP_PAR_bis.txt** is identical to the marker map file provided by the user with the **typ_eval** format, except the last field (on the right) contains 1 for the markers selected by the program and 0 for the markers discarded by the program. It will then be used by the software for Step_2 and Step_3.

Parameter file read by SNP-GWAS at Step 2:

This parameter file for Step 2 has to be prepared by the user, for each of the chromosome he wants to analyze.

NB: the information in red are comments and should not be present in the parameter file!

The information in green are the same (and have to be the same) as in parameter file for Step 1.

```
# Example param file for STEP 1 prg SNP_GWAS      1 line for title
STEPS
2                          ← number of the Step that will be executed (1 for Step 1; 2 for Step 2; 3 for Step3)
EXEC_PATH
/softwares/SNP_GWAS/bin/SNP_GWAS   ← path for binary of the program, needed in Step 2 for Step 3
BIN_DIR
/work/DIR2/Example3/binfiles    ← folder where binary files for LHS-1 & RHS will be written
DATAFILE
/work/DIR2/Example3/FILES/data_file.txt    ← performance file
NUMBER_OF_EFFECTS
1                ← number of non-genetic effects in the model (>= 1), at least 1 for the overall mean
POS_ID_CHAR
6                ← column number of animal Id in performance file: always last column read!
OBSERVATION(S)
4                ← column number of phenotype in performance file
WEIGHT(S)
                 ← column number of phenotype's weight in performance file (leave empty if no weight, i.e. weight=1)
EFFECTS: POSITIONS_IN_DATAFILE NUMBER_OF_LEVELS TYPE_OF_EFFECT   [EFFECT NESTED]
1 1 cross   ← column number of first non-genetic effect in performance file / number of levels / effect type
RESIDUAL_VARIANCE
18.76608     ← residual variance of the trait for a phenotype whose weight is 1= σ_e^2, supposed known
GENETIC_VARIANCE
6.06204      ← genetic variance of the trait = σ_u^2, supposed known
TYP_PAR_FILE
/work/DIR2/Example3/FILES/Markers_50K.raw    ← marker genotype file
FORMAT_TYP_PAR
plink                  ← format type of marker genotype file  (among plink, typ_eval)
MAP_PAR_FILE
/work/DIR2/Example3/FILES/Markers_50K.map    ← marker map file
TYP_GWAS_FILE
/work/DIR2/Example3/FILES/Markers_50K.raw    ← TV genotype/allelic dosage file
FORMAT_TYP_GWAS
plink                 ← format type of TV genotype/allelic dosage file  (among minimac, typ_eval, plink)
NUMCHR_GWAS
14                                          ← chromosome carrying the TV
MAP_GWAS_FILE
/work/DIR2/Example3/FILES/Markers_50K.map          ← TV map file
GWAS_TYPE (add / add_dom)
add                                         ← effect(s) to be considered in the model for TV
SEGM_REM
0                                     ← which markers will be removed form equations during GWAS
COJO
0                ← 0 if no variant fit in the model as fixed effect; 1 if 1 variant fit in the model as fixed effect*
OPTION missing -9999.0                    ← OPTIONS , see below for more details on OPTIONS
OPTION MAF_minimum 0.20
OPTION NbMaxVarGWAS 15000
OPTION CalcVarRes optim
OPTION seuil_test1 2.0
OPTION StepFichOut 100
OPTION SelAutoSNPpar 10000
```

`TYP_GWAS_FILE`

After this keyword is the file containing the genotypes or the allelic dosages for the variants whose effect(s) will be estimated and tested. Only the variant from 1 single chromosome (designed under the **`NUMCHR_GWAS`** keyword below) will be considered). The format for this file will be given below in the **`FORMAT_TYP_GWAS`** section.

NB : **for minimac4 VCF files** : the names of the N VCF files must be like, for example: SEQ_chr21_**1.dose.vcf**, SEQ_chr21_**2.dose.vcf**, …, SEQ_chr21_**N.dose.vcf**, with **1, 2, … N is an unbroken sequence of integers ranging from 1 to N, all the N files must exist !**

In other words, **the N VCF file names** (N can be 1 if only 1 VCF file) **all end with k.dose.vcf** (“**dose.vcf” is obligatory in the current version of the program, so the user has to rename the VCF files accordingly, sorry …**) where k is ne number of the VCF file.

In this case, the user will provide here the common part of the N VCF files, e.g.

/DATA/DIR1/FILES/SEQ_chr21_

And the program will reconstitute the N file names.

`FORMAT_TYP_GWAS`

After this keyword, indicate the kind of genotype/allelic dosage file for the variants whose effect(s) will be estimated and tested.

4 different formats are recognized by the program:

plink:

In this case, the file contains <u>discrete genotypes</u> for the variants, corresponding to the number of allele “2” carried by individuals at a loci. **Unknown genotype has to be coded “5”**. The format is the **.raw** format created by plink software, but **the user has to replace the NA for unknown genotypes by 5 before running SNP_GWAS** (SNP_GWAS reads genotypes as numerics); header line:

NB: FID (family ID) and IID (individual ID) have to be the identical; PAT, MAT, SEX, PHENOTYPE are not used, so dummy values can be used for these variables.

```
FID IID PAT MAT SEX PHENOTYPE F0100190_A D5010001_A F0100220_A RGX1000_A RGX2000_A RGX1001_A …
XXXXX000002130 XXXXX000002130 XXXXX4598225 XXXXX2754989 0 -9 2 0 1 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 0
XXXXX000002131 XXXXX000002131 XXXXX2598225 XXXXX2754989 0 -9 2 0 1 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 0
XXXXX000002132 XXXXX000002132 XXXXX2598225 XXXXX2754989 0 -9 2 0 1 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 0
XXXXX000002133 XXXXX000002133 XXXXX2598225 XXXXX2754989 0 -9 1 0 1 0 0 0 0 0 1 1 0 0 0 1 0 0 2 1 0
XXXXX000002134 XXXXX000002134 XXXXX2598225 XXXXX2754989 0 -9 1 0 2 0 0 0 0 0 0 0 0 0 0 1 0 0 1 1 0
XXXXX000002135 XXXXX000002135 XXXXX2598225 XXXXX2754989 0 -9 1 0 2 0 0 0 0 0 0 0 0 0 0 1 0 0 1 1 0
```

The file can contain the genotypes for variants located on several chromosomes, but <u>only the variants on the chromosome designed under the **`NUMCHR_GWAS`** keyword will be considered in the computations</u>.

The markers in this variant genotype file have to be consistent (= the same) as the marker in the variant map file (see below).

Of course, if the genotype file for markers (`TYPE_PAR_FILE`) is also a **.raw** plink file, the marker set and the variant set, and therefore the marker type file and the variant type file, can differ.

In this file, the markers (column) have to be ordered by ascending chromosome, then in ascending position on the chromosome.

typ_eval:

In this case, the file contains <u>discrete genotypes</u> for the variants, corresponding to the number of allele “2” carried by individuals at a loci: 1 single line per individual, starting with the individual Id, 1 or several <u>spaces (**no tab**)</u>, followed by its genotypes for all variants ordered in ascending position on chromosome, without any separation**. The first position of the genotype of the first variant has to be the same for all individuals, whatever the individual’s Id length**. <u>No header</u>.

```
FRXXX8037618   02121212110222
FRXXX8162      12121211220211
FRXXX001357    21022221110220
FRXXX0035018   11220210112222
```

The file contains <u>only the genotypes for variants located on the chromosome designed under the **`NUMCHR_GWAS`** keyword</u>. Therefore, even with the same density, the marker genotype typ_eval file and the variant genotype typ_eval file are different files.

In this file, the markers (column) have to be ordered by ascending position on the chromosome.
Unknown genotype for an individual at a given variant has to be coded "5".

minimac3: = file format created by minimac version 3 software after imputation.

In this case, the unique file contains allelic dosages for the variants, 1 variant per column. 1 single line per individual, no header:

```
FR0103307315->FR0103307315    DOSE    0.004   0.087   0.002   0.002   0.008   0.199   0.066   0.012   0.005   0.006
FR0103307328->FR0103307328    DOSE    0.004   0.102   0.002   0.002   0.007   0.161   0.051   0.011   0.005   0.005
FR0103307345->FR0103307345    DOSE    0.005   0.090   0.001   0.002   0.007   0.168   0.072   0.012   0.005   0.005
FR0103307732->FR0103307732    DOSE    0.002   0.100   0.001   0.002   0.006   0.153   0.053   0.009   0.004   0.004
FR0103307757->FR0103307757    DOSE    0.004   0.084   0.002   0.002   0.007   0.275   0.064   0.012   0.005   0.006
```

1 line per individual, starting by animal's Id repeated twice and separated by "->", "DOSE", then the variant allelic dosages.

minimac4: = VCF file(s) created by **minimac version 4 software after imputation**. 1 variant per line, 1 animal per column:

```
##fileformat=VCFv4.1
##filedate=2026.1.7
##source=Minimac4.v1.0.2
##contig=<ID=21>
##INFO=<ID=AF,Number=1,Type=Float,Description="Estimated Alternate Allele Frequency">
##INFO=<ID=MAF,Number=1,Type=Float,Description="Estimated Minor Allele Frequency">
##INFO=<ID=R2,Number=1,Type=Float,Description="Estimated Imputation Accuracy (R-square)">
##INFO=<ID=ER2,Number=1,Type=Float,Description="Empirical (Leave-One-Out) R-square (available only f
##INFO=<ID=IMPUTED,Number=0,Type=Flag,Description="Marker was imputed but NOT genotyped">
##INFO=<ID=TYPED,Number=0,Type=Flag,Description="Marker was genotyped AND imputed">
##INFO=<ID=TYPED_ONLY,Number=0,Type=Flag,Description="Marker was genotyped but NOT imputed">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=DS,Number=1,Type=Float,Description="Estimated Alternate Allele Dosage : [P(0/1)+2*P(1/1
##minimac4_Command=minimac4 --refHaps refPanel21.m3vcf --haps genotypes_1
#CHROM  POS     ID          REF   ALT   QUAL  FILTER  INFO    FORMAT  ANIM00001       ANIM00002       ANI
21      12447   21:12447:T:C    T     C     .     PASS    AF=0.10410;MAF=0.10410;R2=0.00000;IMPUTED
21      12450   21:12450:CT:C   CT    C     .     PASS    AF=0.03625;MAF=0.03625;R2=0.00000;IMPUTED
21      12475   21:12475:C:T    C     T     .     PASS    AF=0.02810;MAF=0.02810;R2=0.00000;IMPUTED
21      12547   21:12547:T:C    T     C     .     PASS    AF=0.00342;MAF=0.00342;R2=0.00000;IMPUTED
21      12609   21:12609:T:G    T     G     .     PASS    AF=0.08582;MAF=0.08582;R2=0.00000;IMPUTED
```

In this case, the parameter file for Step #2 includes the following lines:

```
FORMAT_TYP_GWAS
minimac4
NUMBER_VCF
8
```

where NUMBER_VCF is an additional compulsory keyword associated with `FORMAT_TYP_GWAS=minimac4`, followed by an integer (8 in the present example) specifying the number of VCF files to be read. This corresponds to the situation where imputation for the same variants is performed separately for several groups of animals, animals 1 to N1 being in the VCF file #1, animals N1+1 to N2 being in the VCF file #2, …

All VCF files must contain the same variants in the same order!

NUMCHR_GWAS

After this keyword, indicate the Id (numeric) of the chromosome carrying the variants for which we estimate and test the effect(s).

If FORMAT_TYP_GWAS=minimac3, minimac4 or typ_eval : the MAP_GWAS_FILE and TYP_GWAS_FILE files only contain Variants and Genotypes/Allelic Dosages for Variants located on this chromosome;

If FORMAT_TYP_GWAS=plink: the MAP_GWAS_FILE and TYP_GWAS_FILE files can contain Variants located on several chromosomes, but only the Variants located on the chromosome following keyword NUMCHR_GWAS will be considered.

MAP_GWAS_FILE

After this keyword, indicate the map file for the variant present in the genotype/allelic dosage file. ***The map file has to contain the same variants as the genotype/allelic dosage file, in the same order; 1 line per variant.***

If the genotype file contains 1 single chromosome, the map file has to contain the same variants of these unique chromosome.

If the genotype file contains several chromosomes, the map file has to contain the same variants of the same chromosomes.

The map file lines (1 line per marker) are ordered by ascending chromosomes (if several chromosomes) and then by ascending position of the variants on the chromosome.

If FORMAT_TYP_GWAS = **plink**, then the variant map file is the **.map** file created by **plink software**: (col#1 = chromosome; col#2=marker name; col#3=?; col#4 = position on chromosome).

```
1       F0100190        0       776231
1       D5010001        0       904701
1       F0100220        0       907810
1       RGX1000 0       989141
1       RGX2000 0       990551
1       RGX1001 0       990831
1       F0133440        0       993060
…
```

NB: in this case, the user can add a 5th column in the plink .map file to indicate for each variant if it should be considered (1) or excluded (0) when estimating and testing the variant effect(s):

```
1       F0100190        0       776231 1
1       D5010001        0       904701 1
1       F0100220        0       907810 0
1       RGX1000 0       989141 0
1       RGX2000 0       990551 1
1       RGX1001 0       990831 1
1       F0133440        0       993060 0
…
```

If this 5th column is missing, the program considers that all variants have to be considered.

If FORMAT_TYP_GWAS = **typ_eval, minimac3 or minimac4**, then the variant map file contains 4 columns: Col#1 = chromosome (same value for all the variants); col#2 = position of the variant on chromosome (bp); col#3 = order of the variant **within this chromosome** (from 1 to K); col#4 = indicator for each variant telling if it should be considered (1) or excluded (0):

```
1 776231 1 1
1 904701 2 1
1 907810 3 0
1 989141 4 0
1 990551 5 1
1 990831 6 1
1 993060 7 0
…
```

GWAS_TYPE (add / add_dom)

After this keyword, indicate the effects to be estimated and tested in Step_3 for the variant k:

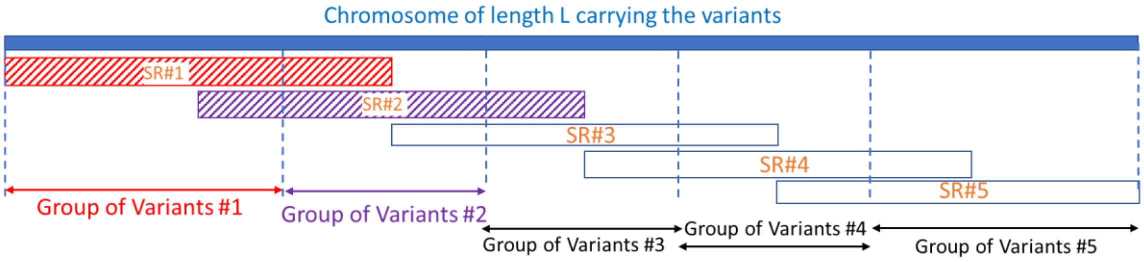
- **add**: only additive effect; the GWAS model is $y = X\beta + Ms + g_k a_k + e$;
- **add_dom**: additive and dominance effect; the GWAS model is $y = X\beta + Ms + g_k a_k + \delta_k d_k + e$;

SEGM_REM

After this keyword, indicate the option selected to remove (or not) markers while estimating and testing the effect(s) of a the variants.

- Put 0 (zero) if you don't want to remove any marker;
- Put 9999999999 (ten consecutive 9) to remove all the markers carried by the same chromosome as the variant's chromosome;

- Put N (positive integer) to remove the markers on a segment of N bp on the variant's chromosome, this segment surrounding the variant.
  NB: in this last case, the chromosome is divided in [2(L/N) – 1] overlapping segments (50% overlap for 2 consecutive segments), where L is the chromosome length; all the markers of a segment are removed from the equations when estimating and testing the effect(s) of the variants located N/4 bp after the beginning of the segment and N/4 bp before the end of the segment, as shown below:



SR#k = segment #k of length N whose markers will be removed when estimating the effects of group #k of variants

  In this case, **the program separates the variants** carried by this chromosme **into** [2(L/N) – 1] **groups** for Step_3, each group containing the variants having the same markers excluded as shown above (e.g. variant in group #3 will have their effects estimated and tested while removing the markers in SR#3).

  If the user specifies to remove all the markers carried by the chromosome (`SEGM_REM=9999999999`) or to keep all the markers carried by the chromosome (`SEGM_REM=0`), then <u>all the variants of the chromosome will be in 1 single group</u> for Step_3.

COJO

This allows the user to specify that they want (put "1" on the line following `COJO` keyword) to include the fixed effect of a chosen variant in the model (additive effect only) while estimating and testing the effect of the variants present in the genotype file, or not (put "0" on the line following `COJO` keyword).

In this case, the GWAS model becomes:

$$y = X\beta + g_{cojo}a_{cojo} + Ms + Z_k b_k + e \,, \tag{3c}$$

where g_{cojo} is the discrete genotype (0, 1, 2, in number of alleles "2") of the additional variant fit in the model, and a_{cojo} is its fixed additive effect, estimated as a regression coefficient whose covariate is g_{cojo}.

In the case you want to fit a variant's additive effect in the model, put "1" on the line following `COJO` keyword and directly after 2 additional lines, as shown below:

```
COJO
1
COJO_FILE
/DATA/Example1/FILES/cojo_var_numb.txt
```

`COJO_FILE` is a keyword (put it exactly like this in parameter file of Step2); on the next line indicate the location and name of the file containing a single value corresponding to the order number (value of field #3 in the `MAP_GWAS_FILE` file) of the variant to be fixed in the model.

Sub-directories and output files created by Step_2:

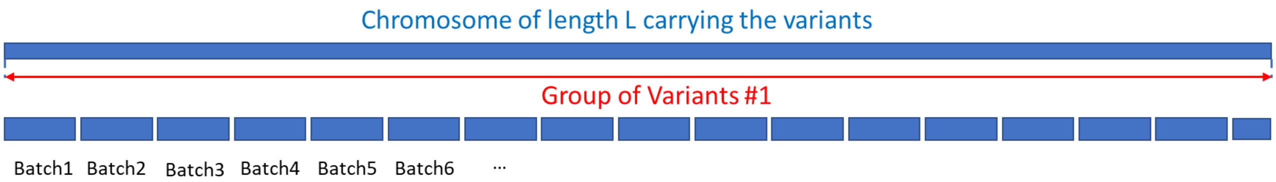
Example #1:

Let's say there are 250,000 variants on chromosome 14, the user decides to keep all the markers for the GWAS (therefore, there is 1 unique variant group for Step_3), and **the user decides to parallelize the GWAS by batches of 15,000 variants** on different CPU of a cluster (this is the meaning of the "**`OPTION NbMaxVarGWAS 15000`**" at the end of the previous steps parameter files).

Then, during Step_2, the program

- creates for Step_3 16 **batches** of 15,000 variants and 1 last batch of 10,000 variants,
- creates 17 sub-folders in the working directory called chr14_gr1/SG1 to chr14_gr1/SG17.

  NB : the number of the chromosome (here 14) is given by the user in the parameter file after the keyword **`NUMCHR_GWAS`**.



Each batch will have its own parameter file, its own binary file containing the genotype/allelic dosage of the 15,000 (or 10,000) variants of the batch (…) created by the software during Step_2:

```
Working_Directory
    ➔ chr14_gr1 = sub-directory created by the program for group #1 of variants
          ▪ SG1   batch 1 sub-folder = variants 1 to 15,000
          ▪ SG2   batch 2 sub-folder = variants 15,001 to 30,000
          ▪ SG3   batch 3 sub-folder = variants 30,001 to 45,000
          ▪ …
          ▪ SG17  batch 17 sub-folder = variants 245,001 to 250,000
```

Example #2:

If the user specified to remove markers from chromosome segments of length N, resulting in separating the variants into **3 groups of variants** containing, respectively, 70,000 , 110,000 and 70,000 variants, and decides to parallelize the GWAS by **batches of 15,000 variants**, the program will create the following sub-folder tree for Stpe_3:

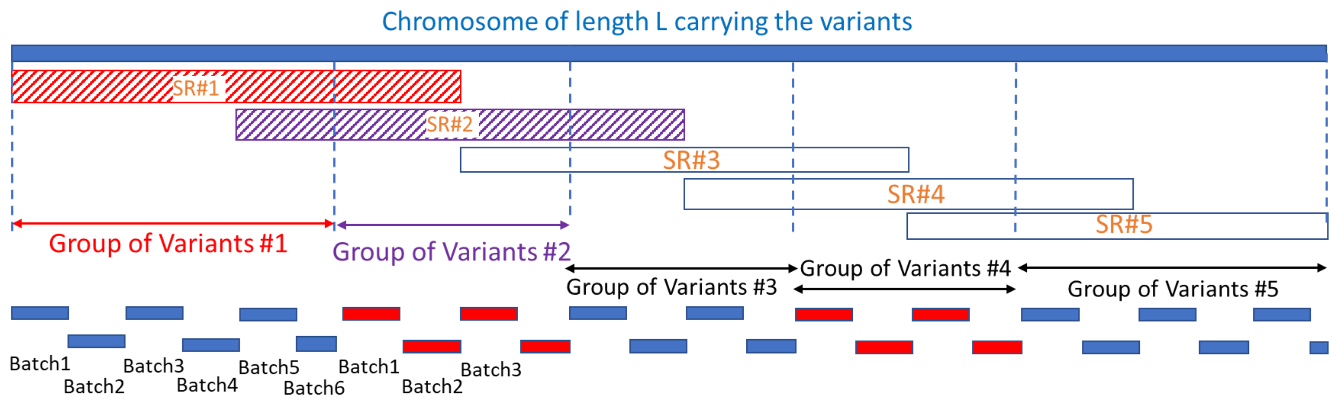
```
Working_Directory

    ➔ chr14_gr1 folder
          ▪ SG1      batch 1 sub-folder = variants 1 to 15,000
          ▪ SG2      batch 2 sub-folder = variants 15,001 to 30,000
          ▪ SG3      batch 3 sub-folder = variants 30,001 to 45,000
          ▪ SG4      batch 4 sub-folder = variants 45,001 to 60,000
          ▪ SG5      batch 5 sub-folder = variants 60,001 to 70,000

    ➔ chr14_gr2 folder
          ▪ SG1      batch 1 sub-folder = variants 70,001 to 85,000
          ▪ SG2      batch 2 sub-folder = variants 85,001 to 110,000
          ▪ SG3      batch 3 sub-folder = variants 110,001 to 125,000
          ▪ …
          ▪ SG8      batch 8 sub-folder = variants 170,001 to 180,000

    ➔ chr14_gr3 folder
          ▪ SG1      batch 1 sub-folder = variants 180,001 to 195,000
          ▪ SG2      batch 2 sub-folder = variants 195,001 to 210,000
          ▪ …
          ▪ SG5      batch 5 sub-folder = variants 240,001 to 250,000
```

In this example, the program creates in total 18 [group x batch] cells for step 3.



In each of these [group x batch] cells, the program creates 3 files during Step_2:

- 1 binary file named **Doses_GrVar.bin** containing the genotypes/allelic dosages (depending on the kind of data read in the variant genotype/allelic dosage file) for the variants in this [group x batch] cell; In this file, 1 line contains the genotypes/allelic dosages for all the individuals for 1 variant, allowing the program to work in Step_3 marker by marker without storing the complete genotype/allelic dosage for all variants in memory.
- 1 parameter file that will be read by the SNP_GWAS software during Step_3 for estimating and testing the effects for the variants in this [group x batch] cell;
- 1 file named **Infos_GrVar.txt** containing 5 numeric variables on 1 single line:

  G B f_marker l_marker batch_size

  Where G is the number of the group;
  B is the number of the batch within group;
  f_marker and l_marker are, respectively, the first and last marker discarded for this group of variants;
  batch_size is the number of variants in this [group x batch] cell.

The program also creates during Step_2 in the working directory an executable file named **lance_part3_allGR_chrN.sh** containing 1 ligne per [group x batch] cell allowing to launch Step_3 for the variants belonging to this cell. Each line has the form:

/path/chrN_grG/SGB/fichpar_part3_GrVar.par > chrN_grG/SGB/SG_G_B.log

NB: if the user requested the **use of "COJO" option** for fitting a designed variant (say variant #12345) in the model:

➔ the executable file is named **lance_part3_allGR_chrNcojo_12345.sh** , and each line in this file will be like:

/path/cojo_12345/chrN_grG/SGB/fichpar_part3_GrVar.par > cojo_12345/chrN_grG/SGB/SG_G_B.log

allowing to run the GWAS successively with several cojo variants without smashing the outputs.

➔ A binary file named **Doses_VarCojo_12345.bin** containing the genotypes/allelic dosages of the cojo variant is created in the /cojo_12345 sub-folder of the working directory, that will be used as the covariates for the fixed regression coefficient for fitting its additive effect in the model during Step_3.

Parameter file read by SNP-GWAS at Step 3:

The parameter files for Step 3 are automatically created by the program during Step 2.
NB: the information in red are comments and should not be present in the parameter file!
The information in green are the same (and <u>have to be the same</u>) as in parameter file for Step 2.

```
# Example param file for STEP 1 prg SNP_GWAS      1 line for title
STEPS
3                                 ← number of the Step that will be executed (1 for Step 1; 2 for Step 2; 3 for Step3)
Chrom_numGR
14  1  2
EXEC_PATH
/softwares/SNP_GWAS/bin/SNP_GWAS  ← path for binary of the program, needed in Step 2 for Step 3
BIN_DIR
/work/DIR2/Example3/binfiles      ← folder where binary files for LHS-1 & RHS will be written
DATAFILE
/work/DIR2/Example3/FILES/data_file.txt    ← performance file
NUMBER_OF_EFFECTS
1                         ← number of non-genetic effects in the model (>= 1), at least 1 for the overall mean
POS_ID_CHAR
6                          ← column number of animal Id in performance file: always last column read!
OBSERVATION(S)
4                       ← column number of phenotype in performance file
WEIGHT(S)
                      ← column number of phenotype's weight in performance file (leave empty if no weight, i.e. weight=1)
EFFECTS: POSITIONS_IN_DATAFILE NUMBER_OF_LEVELS TYPE_OF_EFFECT   [EFFECT NESTED]
1 1 cross             ← column number of first non-genetic effect in performance file / number of levels / effect type
RESIDUAL_VARIANCE
18.76608              ← residual variance of the trait for a phenotype whose weight is 1= σ_e^2, supposed known
GENETIC_VARIANCE
6.06204                 ← genetic variance of the trait = σ_u^2, supposed known
TYP_PAR_FILE
/work/DIR2/Example3/FILES/Markers_50K.raw     ← marker genotype file
FORMAT_TYP_PAR
plink                           ← format type of marker genotype file  (among plink, typ_eval)
MAP_PAR_FILE
/work/DIR2/Example3/FILES/Markers_50K.map     ← marker map file
TYP_GWAS_FILE
/work/DIR2/Example3/FILES/Markers_50K.raw     ← TV genotype/allelic dosage file
FORMAT_TYP_GWAS
plink                         ← format type of TV genotype/allelic dosage file  (among minimac, typ_eval, plink)
NUMCHR_GWAS
14                                               ← chromosome carrying the TV
MAP_GWAS_FILE
/work/DIR2/Example3/FILES/Markers_50K.map               ← TV map file
GWAS_TYPE (add / add_dom)
add                                             ← effect(s) to be considered in the model for TV
SEGM_REM
0                                         ← which markers will be removed form equations during GWAS
COJO
0                       ← 0 if no variant fit in the model as fixed effect; 1 if 1 variant fit in the model as fixed effect*
OPTION missing -9999.0                         ← OPTIONS , see below for more details on OPTIONS
OPTION MAF_minimum 0.20
OPTION NbMaxVarGWAS 15000
OPTION CalcVarRes optim
OPTION seuil_test1 2.0
OPTION StepFichOut 100
OPTION SelAutoSNPpar 10000
```

Two lines have been added in the parameter file, compared to the parameter file for Step_2:

```
Chrom_numGR
14  1  2
```

`Chrom_numGR` is a keyword, followed by 3 integer values:

"14" is the number of the chromosome carrying the variants, as specified in Step_2 after the keyword `NUMCHR_GWAS`;

"1" is the Id of the group of variants, defined by the user's choice for removing markers; here, the user specified "0" for `SEGM_REM` (no marker removed), therefore there is only 1 group of variants for Step_3.

Files read by the program during Step_3:

For each [group x Batch] cell of variants, the SNP_GWAS program reads the following files during Step_3:

- Performance file ;
- Marker genotype file
- Marker map file created by SNP_GWAS during Step_1 tagging the markers to be considered (filtered for MAF, automatic selection if requested, …)
- Binary file containing the genotypes/allelic dosages of the variants (if not minimac4) or the Id of the variants (if minimac4) belonging to this [group x Batch] cell, created in Step_2 /chrN_grK/SGx**/Doses_GrVar.bin** ;
- Variant map file;
- File **/Infos_all_Gr_SG_chrN.txt** created during Step_2
- If minimac4 : files **/vec_DosesUtil_all_Gr_SG_chrN.txt** and **/vec_ordinvdoses_all_Gr_SG_chrN.txt** created during Step_2
- Parameter file /chrN_grK/SGx**/fichpar_part3_GrVar.par** created during Step_2;
- File /chrN_grK/SGx**/Infos_GrVar.txt** created during Step 2.

Output files created by Step_3:

During Step 3, the program SNP_GWAS creates, for each [group G x Batch B] cell of variants for the considered chromosome, 1 or 2 result files, depending on the model considered for the GWAS variants.

1. **If the model considered for the variants is add (only additive effect) or add_dom (additive & dominance effects),** the program creates 1 result file for each cell of variant in the /chrN_grG/SGB sub-folder of the working directory folder (or in the /cojo_NNN/chrN_grG/SGB sub-folder of the working directory folder if the "cojo" functionality was used).
   This file, named **Resultats_chrN_grG.txt**, contains 1 line per variant and the following variables (from left to right) separated by spaces:

   <u>if CalcVarRes=approx</u> (10 columns for "add" model or 13 columns for "add_dom" model)**:**

   - Order of the variant in the considered cell;
   - Initial order of the variant in the variant map file;
   - Mean of allelic dosage / average number of 2 alleles in the sample for the variant;

- Standard deviation of allelic dosage / of 2 alleles in the sample for the variant;
- Indicator (0 /1) of dominance effect in the model for this variant;
- Solution for estimated additive effect for the variant;
- *Solution for the dominance effect for the variant, if considered in the model*;
- Test statistic for the additive effect, calculated with the approximate variance of residuals;
- *Test statistic for the dominance effect, if considered in the model, calculated with the approximate variance of residuals;*
- P-value for additive effect, calculated with the approximate variance of residuals;
- *P-value for dominance effect, if considered in the model, calculated with the approximate variance of residuals;*
- Approximate variance of residuals for the variant;
- The determinant of $Z'_k Z_k$ matrix for add_dom model, to check the estimability of dominance effect for the variant (0.0 for add model).

<u>if CalcVarRes=exact</u> (10 columns for "add" model or 13 columns for "add_dom" model)**:**

- Order of the variant in the considered cell;
- Initial order of the variant in the variant map file;
- Mean of allelic dosage / average number of 2 alleles in the sample for the variant;
- Standard deviation of allelic dosage / of 2 alleles in the sample for the variant;
- Indicator (0 /1) of dominance effect in the model for this variant
- Solution for estimated additive effect for the variant;
- *Solution for the dominance effect for the variant, if considered in the model*;
- Test statistic for the additive effect, calculated with the exact variance of residuals;
- *Test statistic for the dominance effect, if considered in the model, calculated with the exact variance of residuals;*
- P-value for additive effect, calculated with the exact variance of residuals;
- *P-value for dominance effect, if considered in the model, calculated with the exact variance of residuals;*
- exact variance of residuals for the variant;
- The determinant of $Z'_k Z_k$ matrix for add_dom model, to check the estimability of dominance effect for the variant (0.0 for add model).

<u>if CalcVarRes=optim</u> (13 columns for "add" model or 18 columns for "add_dom" model)**:**

- Order of the variant in the considered cell;
- Initial order of the variant in the variant map file;
- Mean of allelic dosage / average number of 2 alleles in the sample for the variant;
- Standard deviation of allelic dosage / of 2 alleles in the sample for the variant;
- Indicator (0 /1) of dominance effect in the model for this variant
- Solution for estimated additive effect for the variant;
- *Solution for the dominance effect for the variant, if considered in the model*;
- Test statistic for the additive effect, calculated with the approximate variance of residuals;
- *Test statistic for the dominance effect, if considered in the model, calculated with the approximate variance of residuals;*
- P-value for additive effect, calculated with the approximate variance of residuals;
- *P-value for dominance effect, if considered in the model, calculated with the approximate variance of residuals;*
- Test statistic for the additive effect, calculated with the exact variance of residuals if it has been calculated for the variant (i.e. if the additive or dominance approximate test-statistic for the variant was bigger than `seuil_test1`);

- *Test statistic for the dominance effect, if considered in the model, calculated with the exact variance of residuals if it has been calculated for the variant;*
- P-value for additive effect, calculated with the exact variance of residuals if it has been calculated for the variant;
- *P-value for dominance effect, if considered in the model, calculated with the exact variance of residuals if it has been calculated for the variant;*
- Approximate variance of residuals for the variant;
- Exact variance of residuals for the variant if it has been calculated for the variant;
- The determinant of $Z'_k Z_k$ matrix for add_dom model, to check the estimability of dominance effect for the variant (0.0 for add model).

Table of possible OPTIONS

Options are specified to the program by adding a line per option at the end of the parameter files for Step 1 and Step 2, in the form: OPTION keyword value, separated by 1 space.

The case (upper or lower case) for the option keywords must be respected.

Options for Step_3 have to be given in the parameter file for step_2; the program will read them during Step_2 and add them in the parameter files it creates for Step_3.

Default value is the value used by the program when no option is specified for a parameter.

Keyword	Step	Possible values and meaning
missing	1, 2 & 3	Value given to the phenotype of an individual in the performance file to make the program understand that it is a missing phenotype; This allows for example to have one single performance file for several traits where some individuals don't have record for some of the traits. Default value = -9999.0
MAF_minimum	1, 2 & 3	MAF value (calculated on GWAS sample) below which a marker is excluded from analyses (does not apply for variants); Default value = 0.01 Constraint: the value specified by the user must be between 0 and 0.5 This minimum MAF only affects markers but has no effect on the variants.
NbMaxVarGWAS	2 (& 3)	Number of variants constituting a batch (=1 job) in Step_3. The last group may contain fewer variants. For example, if the genotype/allelic dosage file for variants contains 1,110,500 variants and the user sets NbMaxVarGWAS=15,000, there will be 74 groups of 15,000 variants and one group of 500 variants. The value of NbMaxVarGWAS should be set so that the CPU time required to execute Step_3 on the NbMaxVarGWAS variants is lower than the CPU time limit allowed for the class of job execution in Step_3. Default value = 5000
CalcVarRes	3	Strategy to calculate the residual variance of the GWAS model, used to calculate the test statistic for the estimated effect(s) of a variant: **approx** = the variance of residuals is calculated by calculating the residuals using exact solutions for the variant effect(s) but using approximate solutions for the non genetic and marker effects (i.e. estimated without any variant in the model) = fastest method; **exact** = the variance of residuals is calculated for each variant with the exact solution for the variant effect(s) and with the exact solutions for the non genetic and marker = longest method; **optim** = for each variant, the variance of residuals is first calculated with the approx method, then with the exact method only for variants having an approximate test statistic larger than **seuil_test1** (see below) for at least one of their effect; the test statistic(s) for this variant is(are) terefore exact. This is a compromise between processing speed and test accuracy. Default value = optim
seuil_test1	3	Value of the test statistic (calculated approximately) for the effect(s) of a variant above which the variance of residuals is recalculated exactly. Default value = 4.0

StepFichOut	3	The results for a variant (estimated effects, test statistics, …) in Step_3 are neither written in the output file on disc variant by variant, nor after the NbMaxVarGWAS have been processed, but every StepFichOut variants - to limit the I/O - so as not to lose all results if execution stops before the last variant in the group Default value = 500
SelAutoSNPpar	1 (& 2 & 3)	OPTION SelAutoSNPpar NNN Option used to ask the program to automatically select the marker with the highest MAF and the most central position in each of the NNN intervals of equal length on all the autosomes. NB: the actual number of markers retained after this automatic selection may be lower than NNN, since some of the NNN interval may only have markers with MAF lower than the default or specified MAF_minimum value. To avoid intervals without marker, set simultaneously a very low value (0) for MAF_minimum if you prefer. Do not set the line "OPTION SelAutoSNPpar NNN" if you do not want to perform this automatic marker selection. In this case, the program will use all markers whose MAF is greater than MAF_minimum and whose 4th column in the marker map file is equal to 1.
MaxMem	2	Maximum amount of memory used to store the variant allelic dosages/genotypes (in real(kind=4)) in Step_2 ; if MaxMem < nb of individuals * nb of variants * 4, the program will process the total number of variants in several pass in Step_2 ; Limits the amount of memory required if the number of individuals is very large. For example, for 300,000 individuals and 1,500,000 variants on a large chromosome, we would need 1.8 TB to store the allelic dosage file in one pass... Default value = 1d+10 (=10Go) Only relevant if variant genotypes/allelic dosages are not in VCF minimac4 files : if minimac4 VCF files, variant allelic dosages/genotypes are not completely read in Step_2.
doses_to_geno	2 & 3	Allows conversion from variant allelic dosage to discrete genotypes if the user provides a variant allelic dosage file but wants to perform GWAS with discrete genotypes. 1 = allelic dosages are converted into discrete genotypes (= number of alleles « 2 » carried by the animal : 0 if allelic dosage <=0.5 ; 1 if 0.5< allelic dosage < 1.5 ; 2 if allelic dosage >= 1.5). Default value = 0
PrintMissGeno	1	0 / 1 : Undetermined genotypes (other than 0, 1, or 2) at kinship markers will be printed in the LOG for a maximum of 100 individuals I OPTION PrintMissGeno 0 (Default) or for all individuals with at least one undetermined genotype if OPTION PrintMissGeno 1.
CentrGenoPar_st1	1	Indicates if the user wants to store the genotypes at kinship markers in STEP_1: - as non-centered integers (0, 1, 2) that will be centered each time they will be used in computations (CentrGenoPar_st1 = 0, default, lower memory usage but higher computing times)

		or - as centered reals (CentrGenoPar_st1 = 1) that can be directly used in computations (higher memory usage but lower computing times) to build the M'WM matrix and M'Wy vector, where M is the matrix of n_s marker centered genotypes for the n individuals Default = 0
CentrGenoPar_st3	3	Indicates if the user wants to store the genotypes at kinship markers in STEP_3 : - as non-centered integers (0, 1, 2) that will be centered each time they will be used in computations (CentrGenoPar_st3 = 0, default, lower memory usage but higher computing times) or - as centered reals (CentrGenoPar_st3 = 1) that can be directly used in computations (higher memory usage but lower computing times). Default = 0
detailed_log	1 2 3	Level of detail printed in the log file: 1 = minimum 2 = more details 3 = even more details
nb_threads_mkl	1 & 3	Integer in [1,10] ; default = 2 Number of threads for mkl matrix or vector operations. To high values for nb_threads_mkl may result in poorer performance.

Parameter file read by Cumul_Res_SG at Step 4:

In step_2 and Step_3, the variants of one chromosome may be separated into several groups (depending on the user's strategy to remove markers on this chromosome) and batches (depending on the number of variants). The program created one result file for each group x batch of variant located in its folder.

The final Step_4 consists **then in gathering all the results for the chromosome in one single file per effect**:

- 1 file for estimated additive effect, named `ADD_Name_for_final_results_chrN`,
- 1 file for estimated dominance effect if considered in the model, named `DOM_Name_for_final_results_chrN`

using the **Cumul_Res_SG** program, where `Name_for_final_results_chrN` is the name chosen by the user for the final results files and specified in the parameter file (see below). This program also calculates -LOG10(P-values) and puts the results in a more readable form than the elementary files.

This parameter file for Step_4 has to be prepared by the user, for each of the chromosome he wants to analyze:

```
Work directory
/work/DIR2/Example3/GWAS
File containing the groups and batches information
Infos_all_Gr_SG_chr14.txt     ← file created by SNP_GWAS at Step_2 in the work directory
Effects considered for the variants(add or add_dom)
add          ← put the same value as in SNP_GWAS parameter file for GWAS_TYPE
Strategy for computing variance of residuals (approx or optim or exact)
optim     ← put the same value as in SNP_GWAS parameter file for CalcVarRes
GWAS Variant map file
/work/DIR2/Example3/FILES/Markers_50K.map    ← fichier carte varGWAS
Format for GWAS Variant MAP file (plink, typ_eval)
plink
Final Result file name
Name_for_final_results_chrN   ← name chosen by user, will be created in work directory
```

NB : for "`Format for GWAS Variant MAP file (plink, typ_eval)`" parameter:

- Indicate **plink** if the variant map file used in steps 2 and 3 is the **.map** file created by **plink software**: (col#1 = chromosome; col#2=marker name; col#3=?; col#4 = position on chromosome), with or without the additional 5th column indicating for each variant if it should be considered (1) or excluded (0) when estimating and testing the variant effect(s):

```
1       F0100190        0       776231
1       D5010001        0       904701
1       F0100220        0       907810
1       RGX1000 0       989141
1       RGX2000 0       990551
1       RGX1001 0       990831
1       F0133440        0       993060
…
```

or

```
1       F0100190        0       776231 1
1       D5010001        0       904701 1
1       F0100220        0       907810 0
1       RGX1000 0       989141 0
1       RGX2000 0       990551 1
1       RGX1001 0       990831 1
1       F0133440        0       993060 0
…
```

- Indicate **typ_eval** if `FORMAT_TYP_GWAS = ` **typ_eval, minimac3 or minimac4** and the .map file used in steps 2 and 3 contains 4 columns:

Col#1 = chromosome (same value for all the variants); col#2 = position of the variant on chromosome (bp); col#3 = order of the variant **within this chromosome** (from 1 to K); col#4 = indicator for each variant telling if it should be considered (1) or excluded (0):

```
1 776231 1 1
1 904701 2 1
1 907810 3 0
1 989141 4 0
1 990551 5 1
1 990831 6 1
1 993060 7 0
…
```

NB : If the user used the **COJO fonctionnality** to fit the fixed additive effect of a variant NN in the GWAS model, Step_2 **created an additional /cojo_NN subfolder** where the /chrM_grk group folders and their /GRj batch sub-folders were created. The elementary result file for each variant group x batch are in the /cojo_NN/chrM_SGk/SGj

Therefore, the parameter file for Step_4 has to be modified as shown below

```
Work directory
/work/DIR2/Example3/GWAS/cojo_NN
File containing the groups and batches information
Infos_all_Gr_SG_chr14.txt ← file created by SNP_GWAS at Step_2 in the work/cojo_NN dir
Effects considered for the variants(add or add_dom)
add          ← put the same value as in SNP_GWAS parameter file for GWAS_TYPE
Strategy for computing variance of residuals (approx or optim or exact)
optim     ← put the same value as in SNP_GWAS parameter file for CalcVarRes
GWAS Variant map file
/work/DIR2/Example3/FILES/Markers_50K.map    ← fichier carte varGWAS
Format for GWAS Variant MAP file (plink, typ_eval)
plink
Final Result file name
Name_for_final_results_chrN   ← name chosen by user, will be created in
                                  the /work/DIR2/Example3/GWAS/cojo_NN directory
```

The program **Cumul_Res_SG** creates final result files containing the following information, from right to left in the files, for each variant of the chromosome (1 line per variant); these files are created in the working directory:

If Variant model = add or add_dom:

- Id of the chromosome;
- Position, in bp, of the Variant on the chromosome;
- Initial order of the variant in the variant map file;
- Number of the Group of Variant containing the Variant in Step_3;
- Number of the Batch within Group of Variant containing the Variant in Step_3;
- Mean of allelic dosage / average number of 2 alleles in the sample for the variant;
- Standard deviation of allelic dosage / of 2 alleles in the sample for the variant;
- Solution for the effect (additive or dominance, depending on the file);
- Test statistic for the considered effect, calculated with the exact variance of residuals if calculated, or calculated with the approximate variance of residuals;
- P-value for the considered effect, calculated with the exact variance of residuals if calculated, or calculated with the approximate variance of residuals;
- -LOG10(P-value) for the considered effect, calculated with the exact variance of residuals if calculated, or calculated with the approximate variance of residuals;
- exact variance of residuals for the variant if calculated, or approximate variance of residuals.

III. Variations of this basic model

1. considering several effects in the non-genetic part of the model

Let's consider the following performance file:

```
1   3   2    124   0.92        2          0.9     FR2345
1   2   1    138   0.98        7          1        FR5432
1   1   3    163   0.99         5         1        FR34334
1   2   5    127   0.74   -9999.0     0        FR15313
1   1   4    156   0.82        12        0.8     FR42211
```

…

field #1 = effect level x_{i0} for overall mean μ (equal to 1 for all individuals)
field #2 = effect level x_{i1} for a fixed categorial effect with 3 levels F1
field #3 = effect level x_{i2} for a fixed categorial effect with 5 levels F2
field #4 = covariate x_{i3} for a regression coefficient RC3
field #5 = covariatex_{i4} for a regression coefficient RC4 nested in effect F1
field #6 = phenotype for trait 1 (individual FR15313 has missing phenotype)
field #7 = weight of phenotype for trait 1
field #8 = alphanumerical Id of the individual.

and let's consider the GWAS model is the following for individual i:

$$y_i = \mu + F2_i + F1_i + x_{i3}RC3 + x_{i3}RC4_{x_{i1}} + M_i s + Z_{ik}b_k + e_i$$

In this example, I've deliberately placed the F2 effect before the F1 effect to clearly show the position of the variables in the performance file below.

Then, the corresponding sections in the parameter files for Step_1, Step_2 (and Step_3) would be:

```
NUMBER_OF_EFFECTS
5               ← 5 non genetic effects: μ, F1, F2, RC3 and RC4
POS_ID_CHAR
8                ← column number of animal Id in performance file: always last column read!
OBSERVATION(S)
6              ← column number of phenotype in performance file
WEIGHT(S)
7             ←column number of phenotype's weight in performance file (leave empty if no weight, i.e. weight=1)
EFFECTS: POSITIONS_IN_DATAFILE NUMBER_OF_LEVELS TYPE_OF_EFFECT  [EFFECT NESTED]
1 1 cross   ← effect 1 is overall mean, whose effect level is in field #1 in performance file
3 5 cross   ←effect 2 is F2, categorial (cross) whose effect level is given in field #3 in perf file and has 5 levels
2 3 cross   ←effect 3 is F1, categorial (cross), whose effect level is given in field #2 in perf file and has 3 levels
4 1 cov    ←effect 4 is RC3, regression coefficient (cov), whose covariate is given in field #4 in perf file & has 1 level
5 3 cov 2  ←effect 5 is RC4, regression coefficient (cov), whose covariate is given in filed #5 in perf file, nested in
             The categorial effect whose level is given in field #2 in perf file, and has 3 level (like F1)
```

2. considering additive and dominance effect for the variant in the GWAS model

To estimate and test both additive and dominance effects for the variants in the GWAS model,
Just specify **add_dom** on the line following the keyword **`GWAS_TYPE`** in the parameter file for Step_2 (this will then be automatically reported in the parameter file for Step_3 created during Step_2).

This model can be used when considering ***discrete genotypes for the variant***, i.e. considering the following model:

$$y = X\beta + Ms + [g_k a_k + \delta_k d_k] + e$$

In this case, g_{ik} is the discrete genotype of individual i at variant k, in number of alleles "2" carried by the individual (0, 1 or 2). Then, δ_{ik} covariate for the dominance effect (automatically determined by the program knowing g_{ik}) is equal to 0 if g_{ik} = 0 or 2, and equal to 1 if g_{ik} = 1.

This model can also be used when considering ***allelic dosages for the variant***, i.e. considering the following model:

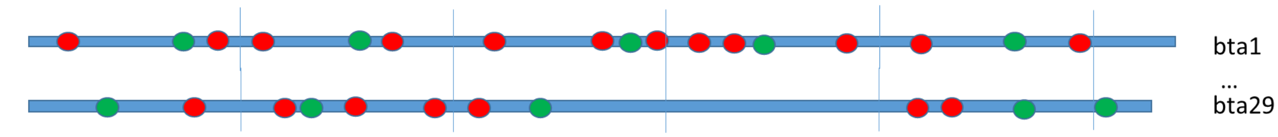
$$y = X\beta + Ms + [\varrho_k a_k + \xi_k d_k] + e$$

In this case, ϱ_{ik} is the allelic dosage (decimal value comprised between 0 and 2) of individual i at variant k, and the covariate ξ_{ik} for the dominance effect (automatically determined by the program knowing ϱ_{ik}) is equal to $||\varrho_{ik} - 1| - 1|$.

3. asking the program to automatically select Markers to model individuals' breeding values

The user has 3 possibilities concerning the set of Markers used to model individuals' breeding values in the GWAS model:

- provide a Marker MAP file ("MAP_PAR_FILE") in which the last column is filled with 1 (or if FORMAT_TYP_PAR=plink, provide a Marker MAP file with only 4 columns, equivalent to a file with a 5th column filled with 1), meaning that all the Markers present in this file whose MAF is higher than the default minimum MAF (0.01) or the minimum MAF specified by user (*via* **`OPTION MAF_minimum 0.XX`**, potentially 0.00) will be considered in the equations;
- provide a Marker MAP file ("MAP_PAR_FILE") in which the last column is filled with 0 for the Markers the User wants to exclude and 1 for the Markers the User wants to include in the equations (NB: a Marker whose last column in MAP file is equal to 1 and whose MAF is lower than the minimum MAF will be excluded anyway);
- ask the program to automatically select Nsegm Markers, evenly spaced and most informative, via the **`OPTION SelAutoSNPpar Nsegm`** (where Nsegm is an integer lower than the total number of Markers in MAP and Genotype files). In this case, the program divides the genome into Nsegm segments of equal size (except the last segment of each chromosome possibly shorter), excludes the Markers whose MAF is lower than minimum MAF, and to retain on each segment the Marker with the highest MAF (i.E. the most informative) and the most central position.
  NB: some segments may have no markers at all in the end, either because they had no markers at all in the MAP file, or because all their markers had a MAF below minimum MAF. The actual number of Markers selected by the program is therefore ≤ Nsegm.



= Marker with highest MAF (> MAFmin) and « in the most central location » in each of the Nsegm segments of equal length distributed throughout the genome

Examples of parameter files

Example #1

Below are the parameter files for GWAS using PLINK genotype files. In this example:

- the Kinship Marker genotype file (keyword `TYP_PAR_FILE`) and GWAS variant genotype file (keyword `TYP_GWAS_FILE`) are the same files;
- 2 effects (the overall mean and 1 regression coefficient) are considered in the environmental part of the model;
- The program selects automatically the best Kinship Marker with MAF > 0.10 in 20000 equal length segments of the genome;
- Approximate test statistics are first calculated for each variant, and then exact test statistic is calculated only for variants having an approximate test statistic > 2.0;
- Batches of 5000 GWAS variants are constituted in Step 2 for Step 3;
- Results are written on disc output file every 100 variant;
- Kinship markers on segments of 10Mb including the GWAS variant will be removed from the equations in STEP 3.

Parameter file for Step 1

```
# parameter file for SNP_GWAS Step_1
STEPS
1
EXEC_PATH
/softwares/GWAS/bin/SNP_GWAS
BIN_DIR
/DATA/milk/TR1/binfiles
DATAFILE
/DATA/Example4/milk/TR1/PerfGWAS_L1_typ_lait_CAR1.txt
NUMBER_OF_EFFECTS
2
POS_ID_CHAR
6
OBSERVATION(S)
4
WEIGHT(S)
3
EFFECTS: POSITIONS_IN_DATAFILE NUMBER_OF_LEVELS TYPE_OF_EFFECT  [EFFECT NESTED]
1 1 cross
2 1 cov
RESIDUAL_VARIANCE
269680
GENETIC_VARIANCE
157390
TYP_PAR_FILE
/DATA/Example4/Markers_50K.raw
FORMAT_TYP_PAR
plink
MAP_PAR_FILE
/DATA/Example4/Markers_50K.map
TYP_GWAS_FILE

FORMAT_TYP_GWAS

NUMCHR_GWAS

MAP_GWAS_FILE

GWAS_TYPE (add / add_dom)
```

```
SEGM_REM

COJO

OPTION missing -9999.0
OPTION MAF_minimum 0.10
OPTION SelAutoSNPpar 20000
OPTION NbMaxVarGWAS 5000
OPTION CalcVarRes optim
OPTION seuil_test1 2.0
OPTION StepFichOut 100
```

Parameter File for Step 2 Chromosome 1

```
# parameter file for SNP_GWAS
STEPS
2
EXEC_PATH
/softwares/GWAS/bin/SNP_GWAS
BIN_DIR
/DATA/milk/TR1/binfiles
DATAFILE
/DATA/Example4/milk/TR1/PerfGWAS_L1_typ_lait_CAR1.txt
NUMBER_OF_EFFECTS
2
POS_ID_CHAR
6
OBSERVATION(S)
4
WEIGHT(S)
3
EFFECTS: POSITIONS_IN_DATAFILE NUMBER_OF_LEVELS TYPE_OF_EFFECT  [EFFECT NESTED]
1 1 cross
2 1 cov
RESIDUAL_VARIANCE
269680
GENETIC_VARIANCE
157390
TYP_PAR_FILE
/DATA/Example4/Markers_50K.raw
FORMAT_TYP_PAR
plink
MAP_PAR_FILE
/DATA/Example4/Markers_50K.map
TYP_GWAS_FILE
/DATA/Example4/Markers_50K.raw
FORMAT_TYP_GWAS
plink
NUMCHR_GWAS
1
MAP_GWAS_FILE
/DATA/Example4/Markers_50K.map
GWAS_TYPE (add / add_dom)
add_dom
SEGM_REM
10000000
COJO
0
OPTION missing -9999.0
OPTION MAF_minimum 0.10
OPTION SelAutoSNPpar 20000
OPTION NbMaxVarGWAS 5000
OPTION CalcVarRes optim
OPTION seuil_test1 2.0
OPTION StepFichOut 100
```

Parameter File for Step 3 Chromosome 1 Group 1 Batch 1 automatically generated in Step 2

```
# parameter file for SNP_GWAS
STEPS
3
Chrom_numGR
1  1  1
EXEC_PATH
/softwares/GWAS/bin/SNP_GWAS
BIN_DIR
/DATA/milk/TR1/binfiles
DATAFILE
/DATA/Example4/milk/TR1/PerfGWAS_L1_typ_lait_CAR1.txt
NUMBER_OF_EFFECTS
2
POS_ID_CHAR
6
OBSERVATION(S)
4
WEIGHT(S)
3
EFFECTS: POSITIONS_IN_DATAFILE NUMBER_OF_LEVELS TYPE_OF_EFFECT  [EFFECT NESTED]
1 1 cross
2 1 cov
RESIDUAL_VARIANCE
269680
GENETIC_VARIANCE
157390
TYP_PAR_FILE
/DATA/Example4/Markers_50K.raw
FORMAT_TYP_PAR
plink
MAP_PAR_FILE
/DATA/Example4/Markers_50K.map
TYP_GWAS_FILE
/DATA/Example4/Markers_50K.raw
FORMAT_TYP_GWAS
plink
NUMCHR_GWAS
1
MAP_GWAS_FILE
/DATA/Example4/Markers_50K.map
GWAS_TYPE (add / add_dom)
add_dom
SEGM_REM
10000000
COJO
0
OPTION missing -9999.0
OPTION MAF_minimum 0.10
OPTION SelAutoSNPpar 20000
OPTION NbMaxVarGWAS 5000
OPTION CalcVarRes optim
OPTION seuil_test1 2.0
OPTION StepFichOut 100
```

Parameter File for Cumul_Res_SG Chromosome 1

```
Work Directory:
/DATA/milk/TR1/GWAS
File containng the Groups and Batches information
Infos_all_Gr_SG_chr1.txt
Effects considered for the GWAS Variants (add / add_dom)
add_dom
Strategy for computing variance of residuals (approx  optim  exact)
optim
GWAS Variant MAP file
/DATA/Example4/Markers_50K.map
Format for GWAS Variant MAP file (plink, typ_eval)
plink
Final Result file name
Final_Res_milk_tr1_chr1_Excl10Mb_modeladddom.txt
```

Example #2

Below are the parameter files for GWAS using 8 minimac4 VCF files containing the allelic dosages of the GWAS variants. Each of the 8 VCF files contains the same GWAS variants with the same ordering, each file contains allelic dosages for different animals (e.g. VCF file #1 contains animals 1 to 1998, VCF file #2 contains animals 19999 to 39997, …).

The names of the 8 VCF files are:
/…/SEQ_r46_chr21_**1**.dose.vcf
/…/SEQ_r46_chr21_**2**.dose.vcf
/…/SEQ_r46_chr21_**3**.dose.vcf
…
/…/SEQ_r46_chr21_**8**.dose.vcf
Only the common prefix of the VCF file names (/…`/SEQ_r46_chr21_`) and the number of VCF files (8) are specified in the parameter file for STEP 2; the program will automatically add the **imposed suffix** **".dose.vcf"** to the end of file name during Steps 2 and 3.

- 1 effect (the overall mean) is considered in the environmental part of the model;
- The user indicates the markers to be considered for kinship in the 4th column of file `TYP_PAR_FILE` `/work/DIR1/Example2/kinship_genotypes` (i.e. no `SelAutoSNPpar` option) with a "1";
- Approximate test statistics are first calculated for each variant, and then exact test statistic is calculated only for variants having an approximate test statistic > 2.5;
- Batches of 7500 GWAS variants are constituted in Step 2 for Step 3;
- Results are written on disc output file every 100 variant;
- Kinship markers on segments of 10Mb including the GWAS variant will be removed from the equations in STEP 3.

Parameter file for Step 1

```
# parameter file for SNP_GWAS Step_1
STEPS
1
EXEC_PATH
/softwares/SNP_GWAS/bin/SNP_GWAS
BIN_DIR
/work/DIR1/Example2/Excl10Mb_add/binfiles
DATAFILE
/work/DIR1/Example2/data_file
NUMBER_OF_EFFECTS
1
POS_ID_CHAR
3
OBSERVATION(S)
2
WEIGHT(S)

EFFECTS: POSITIONS_IN_DATAFILE NUMBER_OF_LEVELS TYPE_OF_EFFECT  [EFFECT NESTED]
1 1 cross
RESIDUAL_VARIANCE
5358
GENETIC_VARIANCE
342
TYP_PAR_FILE
/work/DIR1/Example2/kinship_genotypes
FORMAT_TYP_PAR
typ_eval
MAP_PAR_FILE
/work/DIR1/Example2/kinship_map
```

```
TYP_GWAS_FILE

FORMAT_TYP_GWAS

NUMCHR_GWAS

MAP_GWAS_FILE

GWAS_TYPE (add / add_dom)

SEGM_REM

COJO

OPTION missing -9999.0
OPTION MAF_minimum 0.0
OPTION NbMaxVarGWAS 5000
OPTION CalcVarRes optim
OPTION seuil_test1 2.0
OPTION StepFichOut 100
OPTION CentrGenoPar_st1 1
OPTION CentrGenoPar_st3 1
```

Parameter File for Step 2 Chromosome 21

```
# parameter file for SNP_GWAS
STEPS
2
EXEC_PATH
/softwares/SNP_GWAS/bin/SNP_GWAS
BIN_DIR
/work/DIR1/Example2/Excl10Mb_add/binfiles
DATAFILE
/work/DIR1/Example2/data_file
NUMBER_OF_EFFECTS
1
POS_ID_CHAR
3
OBSERVATION(S)
2
WEIGHT(S)

EFFECTS: POSITIONS_IN_DATAFILE NUMBER_OF_LEVELS TYPE_OF_EFFECT  [EFFECT NESTED]
1 1 cross
RESIDUAL_VARIANCE
5358
GENETIC_VARIANCE
342
TYP_PAR_FILE
/work/DIR1/Example2/kinship_genotypes
FORMAT_TYP_PAR
typ_eval
MAP_PAR_FILE
/work/DIR1/Example2/kinship_map
TYP_GWAS_FILE
/work/DIR1/Example2/SEQ_r46_chr21_
FORMAT_TYP_GWAS
minimac4
NUMBER_VCF
8
NUMCHR_GWAS
21
MAP_GWAS_FILE
/work/DIR1/Example2/GWASvar_map
GWAS_TYPE (add / add_dom)
add
SEGM_REM
10000000
COJO
0
OPTION missing -9999.0
OPTION MAF_minimum 0.0
OPTION NbMaxVarGWAS 7500
OPTION CalcVarRes optim
OPTION seuil_test1 2.5
OPTION StepFichOut 100
OPTION CentrGenoPar_st1 1
OPTION CentrGenoPar_st3 1
```

Parameter File for Step 3 Chromosome 21 Group 3 Batch 4

```
# parameter file for SNP_GWAS
STEPS
3
Chrom_numGR
21  3  4
EXEC_PATH
/softwares/SNP_GWAS/bin/SNP_GWAS
BIN_DIR
/work/DIR1/Example2/Excl10Mb_add_boucle/binfiles
DATAFILE
/work/DIR1/Example2/data_file
NUMBER_OF_EFFECTS
1
POS_ID_CHAR
3
OBSERVATION(S)
2
WEIGHT(S)

EFFECTS: POSITIONS_IN_DATAFILE NUMBER_OF_LEVELS TYPE_OF_EFFECT  [EFFECT NESTED]
1 1 cross
RESIDUAL_VARIANCE
5358
GENETIC_VARIANCE
342
TYP_PAR_FILE
/work/DIR1/Example2/kinship_genotypes
FORMAT_TYP_PAR
typ_eval
MAP_PAR_FILE
/work/DIR1/Example2/kinship_map
TYP_GWAS_FILE
/work/DIR1/Example2/SEQ_r46_chr21_
FORMAT_TYP_GWAS
minimac4
NUMBER_VCF
8
NUMCHR_GWAS
21
MAP_GWAS_FILE
/work/DIR1/Example2/GWASvar_map
GWAS_TYPE (add / add_dom)
add
SEGM_REM
10000000
COJO
0
OPTION missing -9999.0
OPTION MAF_minimum 0.0
OPTION NbMaxVarGWAS 7500
OPTION CalcVarRes optim
OPTION seuil_test1 2.0
OPTION StepFichOut 100
OPTION CentrGenoPar_st1 1
OPTION CentrGenoPar_st3 1
```

Parameter File for Cumul_Res_SG Chromosome 21

```
Work Directory:
/work/DIR1/Example2/Excl10Mb_add_boucle/GWAS
File containing the Groups and Batches information
Infos_all_Gr_SG_chr21.txt
Effects considered for the GWAS Variants (add / add_dom)
add
Strategy for computing variance of residuals (approx  optim  exact)
optim
GWAS Variant MAP file
/work/DIR1/Example2/GWASvar_map
Format for GWAS Variant MAP file (plink, typ_eval)
typ_eval
Final Result file name
Res_Final_chr21_Excl10Mb_add.txt
```